

Abhinav Prakash

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SUMMARY

- Software Development Engineer in Test (SDET) with **5 years** of experience architecting test infrastructure, CI/CD pipelines, and shared engineering utilities for complex hardware-software ecosystems.
- Strong programming expertise in Python and Java, specializing in developing internal developer tools to accelerate system setups, test execution, and deployment validation.
- Deep expertise in DevSecOps and network monitoring (IIT Roorkee PGP), integrating tools like Burp Suite and Wireshark directly into automated pipelines to ensure secure execution environments.

EXPERIENCE

Senior Software QA Engineer

Apr 2025 – Present

ChargePoint, Gurugram, India

- Served as the technical project QA lead, driving infrastructure architecture and automation strategies for critical EV charging releases without direct people management, ensuring highly reliable hardware-software integration.
- Spearheaded the development of "DevUtils" and "ARM" (Automation Resource Management), building Python, FastAPI, and Streamlit-based shared utilities for system setups, deep log analysis, and test orchestration, reducing defect triage time by 30%.
- Engineered and integrated modular automation frameworks into Jenkins CI/CD pipelines utilizing Docker, achieving **40% faster execution** in deployment cycles.
- Validated complex embedded systems logic, including power-sharing algorithms and secure firmware updates over OCPP protocols, mirroring high-reliability OS requirements.

Software QA Engineer

Apr 2022 – Apr 2025

ChargePoint, Gurugram, India

- Acted as the first level of debugging for developer teams by developing custom Python and Shell scripts to continuously parse Unix system logs and query RDBMS tables, cutting defect triage time by 30%.
- Designed scalable test frameworks from scratch using Python and Playwright, **reducing manual infrastructure setup and test effort by 25%**.
- Leveraged cybersecurity expertise to embed fundamental security checks, utilizing Wireshark and Burp Suite to monitor network traffic and validate API security during automated runs.
- Executed comprehensive backend testing and automated high-priority REST API workflows, ensuring fault-tolerant communication between edge devices and cloud services.

Programmer Analyst Trainee

Feb 2021 – Mar 2022

Cognizant

- Automated critical UI test cases for CVS Health projects using Selenium with Java, **improving pipeline reliability and test coverage**.
- Contributed to framework enhancements and execution pipelines, supporting Agile delivery cycles and improving overall CI efficiency.

EDUCATION

PGP, Cyber Security

Jul 2023 – Mar 2024

Indian Institute of Technology, Roorkee

B.Tech, Electronics & Communications Engineering

2017 – 2021

Jaypee Institute Of Information Technology

SKILLS

Frameworks & Programming: Python, Java, Pytest, Playwright, Selenium, FastAPI, Streamlit

DevOps & Infrastructure: Unix, Shell Scripting, Linux, Jenkins, Docker, CI/CD Pipelines

Databases & Security: RDBMS (SQL/PostgreSQL), Wireshark, Burp Suite, Network Monitoring

Protocols & APIs: REST API Testing, Postman, Swagger, OCPP 1.6/2.0 (Hardware Integration)

CERTIFICATIONS

Learning Jenkins — 5 Common Test Failures — Data Structures and Algorithms – Coding Blocks